VOLTLAB / FEBRO HERDYANTO

Industrial Electrical Learning Guide

From electrical fundamentals to industrial control and automation.

Learning sequence

Start with fundamentals, practice DC and AC circuits, study protection and motor control, then continue to panel design, troubleshooting and automation.

Safety boundary: this guide is for learning. VoltLab uses ideal models and does not approve designs, select protection or replace work procedures. Use manufacturer data, applicable standards, qualified personnel and energy-isolation procedures before real work.

01 / BEGINNER

Electrical fundamentals

Prerequisite: No prior experience required.

Module outcome: Distinguish voltage, current, resistance and power; read simple series and parallel paths.

Lessons

Voltage

Voltage is potential difference between two points. Measure across a component, not from one terminal alone.

Practice: Use a meter in parallel across the load.

Current

Current flows only when a complete path exists. An open switch interrupts that path.

Practice: Identify the source, load and return before tracing current.

Resistance and Ohm’s Law

For a linear resistor, V = I × R. At fixed voltage, higher resistance means lower current.

Practice: Calculate current through 96 Ω at 24 V.

Power

For DC, P = V × I. Real AC active power also depends on power factor.

Practice: Estimate DC load power, then state what the ideal model omits.

Series and parallel paths

Series loads share current; parallel branches share voltage. Branch currents add at the source.

Practice: Compare two resistors in series and in parallel.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

02 / BEGINNER

DC circuits

Prerequisite: Electrical fundamentals.

Module outcome: Wire DC sources and loads with correct polarity and a complete return path.

Lessons

Battery and supply

A DC source has positive and negative terminals. VoltLab models an ideal source and does not include internal resistance.

Practice: Label positive, load and return on a simple circuit.

Switch and continuity

A closed contact conducts; an open contact breaks continuity. ON alone does not prove the load has a return.

Practice: Open one connection and locate the first lost path.

LED and resistor

An LED needs current limiting. A first estimate is R = (Vs − Vf) / I. Verify power and standard values for real work.

Practice: Estimate a series resistor for a 12 V source, 2 V LED, 20 mA target.

Relay and polarity

A relay coil controls separate contacts. Check A1–A2 rating and AC/DC type; preserve polarity for sensitive DC loads.

Practice: Operate a relay coil and observe NO and NC contact state.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

03 / BEGINNER

AC fundamentals

Prerequisite: Electrical fundamentals.

Module outcome: Identify L, N and PE; understand RMS voltage, frequency and phase relationships.

Lessons

Live, neutral and protective earth

L is the active conductor. N provides the normal return in a single-phase circuit. PE is a protective conductor, not a normal load return.

Practice: Trace L and N through a 230 V load. Keep PE function separate.

RMS and frequency

AC values are commonly expressed as RMS. Frequency in hertz counts cycles per second. VoltLab uses an ideal phasor model.

Practice: Explain what a 50 Hz rating means.

Three phase

L1, L2 and L3 are separated by 120°. For a balanced system, VLL = √3 × VLN. Phase sequence affects motor direction.

Practice: Relate 400 V line-to-line to approximately 231 V phase-to-neutral.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

04 / INTERMEDIATE

Protection devices

Prerequisite: AC/DC fundamentals and terminal identification.

Module outcome: Differentiate switching, overcurrent, short-circuit and motor overload functions.

Lessons

Fuse, MCB and MCCB

Each device has ratings, breaking capacity and application limits. The simulator does not calculate physical time-current curves or selectivity.

Practice: Compare device nameplate data with the circuit design, not appearance alone.

Overload relay

The training model opens NC 95–96 during a manual trip. An overload relay does not replace short-circuit protection.

Practice: Trace 95–96 in series with the contactor coil.

Protective earth

PE connects exposed conductive parts into the protective earthing system. It is not a normal current return.

Practice: Explain why PE and neutral must not be treated as interchangeable.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

05 / INTERMEDIATE

Motor control

Prerequisite: DC circuits and protection-device functions.

Module outcome: Build a contactor control path with START/STOP, self holding and overload interruption.

Lessons

Contactor and relay

A1–A2 is the coil. Main contacts switch the load; auxiliary 13–14 NO and 21–22 NC support control logic.

Practice: Separate the coil path from the three-phase power path.

START / STOP and self holding

STOP NC is in series. START NO momentarily energizes the coil. Auxiliary NO 13–14 is wired in parallel with START to hold the coil.

Practice: Press START, release it, then press STOP. Observe contactor state.

DOL starter

A DOL path uses upstream protection, contactor main poles and overload before the motor. Control interruption must drop the coil.

Practice: Trace each phase and the control loop separately.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

06 / ADVANCED

Industrial panel design

Prerequisite: Motor control and panel component identification.

Module outcome: Arrange devices and wiring for legibility, maintenance access and traceability.

Lessons

DIN rail and mounting

Mount only compatible modular devices. Confirm rail type, latch engagement, spacing and service access.

Practice: Explain the key point for DIN rail and mounting and test it on a labelled practice circuit.

Wire duct and segregation

Route conductors by function. Follow enclosure fill, bend radius, separation and thermal requirements from approved design and manufacturer data.

Practice: Explain the key point for Wire duct and segregation and test it on a labelled practice circuit.

Power and control routing

Keep the power path and control path distinguishable in the drawing. Label every wire and terminal to match the schematic.

Practice: Explain the key point for Power and control routing and test it on a labelled practice circuit.

Terminal blocks and BOM

Terminal blocks join field cables to panel wiring. A BOM lists tag, description, rating, quantity and source reference.

Practice: Explain the key point for Terminal blocks and BOM and test it on a labelled practice circuit.

Cable sizing overview

Cable sizing depends on load, installation method, temperature, length, voltage drop, protection and local standards. Obtain engineering approval.

Practice: Explain the key point for Cable sizing overview and test it on a labelled practice circuit.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

07 / ADVANCED

Troubleshooting

Prerequisite: AC fundamentals, protection and motor control.

Module outcome: Use symptoms, drawings and measurements to isolate a fault without guessing.

Lessons

Open circuit

Trace source, protective device, control contacts, load and return. Find the first point where the expected path is absent.

Practice: Explain the key point for Open circuit and test it on a labelled practice circuit.

Wrong NO / NC

NO is open and NC is closed in the normal, unoperated state. Check terminal numbers and the device diagram.

Practice: Explain the key point for Wrong NO / NC and test it on a labelled practice circuit.

Coil and overload faults

Measure A1–A2 under the command. Check rating, supply type, return path and 95–96 continuity. Follow approved safe-work procedures.

Practice: Explain the key point for Coil and overload faults and test it on a labelled practice circuit.

Phase loss

Trace L1/L2/L3 through each pole and overload to U/V/W. This ideal model does not calculate motor heating.

Practice: Explain the key point for Phase loss and test it on a labelled practice circuit.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

Industrial automation

Prerequisite: Motor control, troubleshooting and instrumentation basics.

Module outcome: Understand the signal chain between sensors, PLC inputs, program logic and outputs.

Lessons

PLC scan and digital input

A PLC reads inputs, executes logic and updates outputs. Check supply, common, input voltage and sinking/sourcing type.

Practice: Explain the key point for PLC scan and digital input and test it on a labelled practice circuit.

Inductive and photoelectric sensors

Check sensing method, range, target, PNP/NPN output, supply and pinout against the manufacturer sheet.

Practice: Explain the key point for Inductive and photoelectric sensors and test it on a labelled practice circuit.

Timer, interlocks and VFD

A timer changes contacts over time. A VFD controls motor speed by varying output frequency. Use approved parameter and interlock documentation.

Practice: Explain the key point for Timer, interlocks and VFD and test it on a labelled practice circuit.

Commissioning records

Verify drawings, device ratings, terminals and test results against the approved procedure. Record deviations and stop if a safety condition is uncertain.

Practice: Explain the key point for Commissioning records and test it on a labelled practice circuit.

Self-check: can you explain the terminal path, relevant ratings, normal state and simulation limits before calling the circuit ready?

Working reference

Use terminal designations from the device data sheet in circuit drawings. Pre-energization checks must follow applicable installation standards and procedures. VoltLab material does not select conductor sizes or breaker ratings for real projects.

Credits

VoltLab by Febro Herdyanto

Independent educational material. Illustrations and simulations are simplified to explain foundational concepts.